

HARIPRIYA NAIR

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SUMMARY

Quantitative finance professional transitioning into research, with expertise in machine learning, statistical modeling, financial risk, and derivatives. M.S. in Quantitative & Computational Finance from Georgia Tech, with professional experience across hedge fund, trading firm, and financial exchange.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

Master of Science in Quantitative and Computational Finance

Atlanta, GA

December 2019

- **Key Courses:** Management of Fin Instt, Derivatives, Fixed Income, Probabilistic Models, Financial & Stochastic Optimization

NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA

Bachelor of Technology in Electrical and Electronics Engineering

Mangalore, India

May 2016

- **Key Courses:** Probability Theory, Linear Algebra, Engineering Mathematics, Engineering Economics

RESEARCH INTERESTS

Machine Learning & AI • Financial Markets • Market Microstructure • HFT • Quantitative Finance • Financial Risk

SELECTED PROJECTS

Firmwide VaR Modeling & Cross-Asset Risk Aggregation (Geneva Trading)

- Led quantitative research for a Firm-wide VaR framework across global futures, designing a four-layer architecture spanning market-data ingestion, intraday risk computation, cross-asset aggregation, and downstream model interfaces.
- Developed and evaluated book-specific statistical risk models across Power, STIR, FX/stat-arb, oil, energy, and freight, selecting methodologies based on the characteristics of each strategy and underlying markets.
- Investigated statistical dependence and cross-asset aggregation, comparing copula and filtered historical simulation approaches & implementing a t-copula framework with rolling static correlations to aggregate heterogeneous risk distributions into a unified, decomposable VaR measure.
- Built the computational architecture for intraday, parallelized VaR estimation engine and AWS-based API integration with internal proprietary tools, dashboards, excel plugins and trading infrastructure.

Market Regime Modeling & Strategy Allocation (Mission Crest)

- Developed probabilistic market-regime models combining unsupervised learning, supervised ML, change-point detection, and statistical heuristics to characterize evolving market environments.
- Constructed macroeconomic risk factors and predictive features from market and publicly available macro data; estimated the probability of transitions between regimes.
- Translated regime forecasts into dynamic capital-allocation recommendations for rotating capital across strategies based on current conditions and the predicted market regime.

QCF Capstone: Study of Downside Risk Measure for Portfolio Optimization (Based on Jarrow-Xhao)

- Investigated portfolio optimization under downside risk and jump-diffusion dynamics, extending mean-variance frameworks.
- Applied statistical jump-detection methods, Monte Carlo simulation to model discontinuous equity returns
- Developed and backtested alternative downside-risk optimization strategies, examining the sensitivity of portfolio behavior to model assumptions and estimated parameters.

PROFESSIONAL EXPERIENCE

GENEVA TRADING USA LLC

Senior Quantitative Risk Analyst

Chicago, USA

November 2025–Present

- Lead quantitative research and development for the firmwide Value-at-Risk model: designing statistical models, time-series methodologies, and cross-asset aggregation approaches across global futures: FX, equity, rates, power, crypto and energy books.
- Designed and deployed a real-time internal CME GC2 credit monitoring platform aggregating live orders and fills across accounts, contracts, and asset classes; used Claude-assisted development to accelerate code generation, debugging, and iteration while independently designing, validating, and productionizing the platform. The system enabled proactive management of order exposures across all books and helped prevent credit-related order rejects, particularly during volatile roll periods.
- Managed and provided critical enhancements to a real-time CME EMT, Eurex, JPX messaging surveillance system calculating real-time Order-to-Trade Ratio (OTR) and Excessive System Usage metrics from live order and execution data; Enhancements include automated trader-facing alerts and throttling capabilities to proactively manage messaging load across exchanges' elec-

tronic trading platforms. Incorporated product-level liquidity profiles into dynamic alert thresholds, enabling earlier intervention for illiquid products while allowing greater messaging activity in liquid markets.

- Supported integration of Kalshi market data and trading workflows into Geneva's risk infrastructure, developing exposure monitoring, position aggregation, and limit-management capabilities for potential market-making expansion.
- Provided live trading oversight: real-time, first-line support and monitoring of live trading activity, rapidly identifying and escalating risk exposures, breaches, or system anomalies
- Market Analysis: Interpreted complex trading behavior and market movements to assess potential risk implications and help continuously refine model parameters.
- Designed and automated an EOD P&L reconciliation pipeline for a highly manual, time-sensitive workflow, reducing daily processing time from ~3 hours to 15 minutes.

MISSION CREST CAPITAL MANAGEMENT, LLC

Chicago, USA

Director, Quantitative Research

March 2022–September 2025

- Conducted quantitative research on cross-asset portfolios spanning rates, commodities, fx, and equity index futures markets, evaluating strategy behavior, signal stability, risk drivers, diversification, and performance across different market regimes.
- Designed a prospective strategy evaluation framework combining PCA, volatility-targeted optimization, and market-regime analysis to identify complementary strategies and support systematic capital allocation decisions.
- Developed macro and scenario-based models using systematic technical indicators, volatility modeling, and historical simulation to assess portfolio behavior across changing market regimes and stress scenarios.
- Built trade- and portfolio-level quantitative analyses for the CIO and senior investment managers, evaluating drawdowns, risk concentrations, performance attribution, and strategy-specific behavior to inform investment decisions.
- Built a real-time Python-based P&L and risk analytics platform for intraday monitoring of trades/exposures, performance attribution, and guideline constraints across the firm using Bloomberg HAPI pricing feed.
- Automated weekly quantitative research and reporting workflows covering P&L attribution, stress-test impacts, risk-factor decomposition, and position-level analytics, reducing manual effort by 80%.

INTERCONTINENTAL EXCHANGE

Atlanta, USA

Senior Quantitative Engineer

February 2022–March 2022

Quantitative Engineer

February 2020–January 2022

- Developed and validated Python-based quantitative models for derivatives pricing, margin, stochastic volatility surfaces, and risk calculations across Interest Rate, Equity Index, Commodity, and Equity Index derivatives.
- Built pricing engines for American and European options, spot and index derivatives, including stochastic models supporting model validation and margin methodologies.
- Developed Python-based anomaly detection and arbitrage-checking tools for option and linear market data, identifying potential pricing and data inconsistencies.
- Developed quantitative models and pricing infrastructure for commodity derivatives & portfolio margining on NGX platform.
- Collaborated with quantitative research and software engineering teams to translate mathematical models into production code.

BANK OF AMERICA

Atlanta, USA

Quantitative Analyst Intern, Wholesale Loss Forecasting

June 2019–August 2019

- Analyzed CECL model outputs using ML models to identify and quantify key drivers of expected credit loss across regional, industry, and business-line portfolios; selected stepwise regression as the final approach for the optimal forecast combination model.
- Implemented the selected model in R and developed an R Shiny dashboard for interactive, real-time visualization of model results and driver contributions, using MAE and RMSE to quantify the incremental impact of individual drivers.

TECHNICAL SKILLS

Programming: Python, C++, KDB/Q, SQL, R **Research / Engineering:** Git, CI/CD, Anaconda, PyCharm, Dash, Streamlit

ML & Statistics: scikit-learn, NumPy, Pandas, SciPy, Time-Series Analysis, PCA, Regime Analysis, Volatility Modeling

Risk & Quantitative Finance: VaR, Stress Testing, Copulas, ARCH models, FHS, Monte Carlo, Derivatives Pricing, Fixed Income, FX, Market Microstructure, Portfolio Construction

Optimization & Computation: Statistical Modeling, Optimization, Numerical Methods, Multithreading

Certifications & Coursework: CFA Level II; Machine Learning, Bayesian Statistics, Financial Markets, Game Theory (Ongoing)

ACADEMIC & PROFESSIONAL SERVICE

- **QCF Alumni Board Member, Georgia Tech** — Served on the inaugural alumni panel, supported national program outreach, and established a Georgia Tech–Geneva Trading connection that facilitated a company campus visit.
- **AI Committee Member, Mission Crest Capital** — Contributed to internal evaluation and adoption of AI/ML applications for quantitative research and investment workflows.
- **Head Teaching Assistant, Data Analysis in Business (MGT 6203), Georgia Tech** — Led a team of 16 TAs, coordinating coursework, grading, and student support while working directly with Prof. Frederic Bien on course delivery.